



BikeSpace

Digital tools to improve bicycle parking in Toronto

Civic Tech Brampton, March 19, 2026

About BikeSpace



Volunteer party January 2025

Jul 2017: First hacknight at Civic Tech TO

Jul 2018: Public launch

Mid-2021: 🦴

Summer 2023: App relaunched

Winter 2024 : Dashboard relaunched

Fall 2024 : Front-end unified

May 2025: New parking map launched

Myrna's Commute

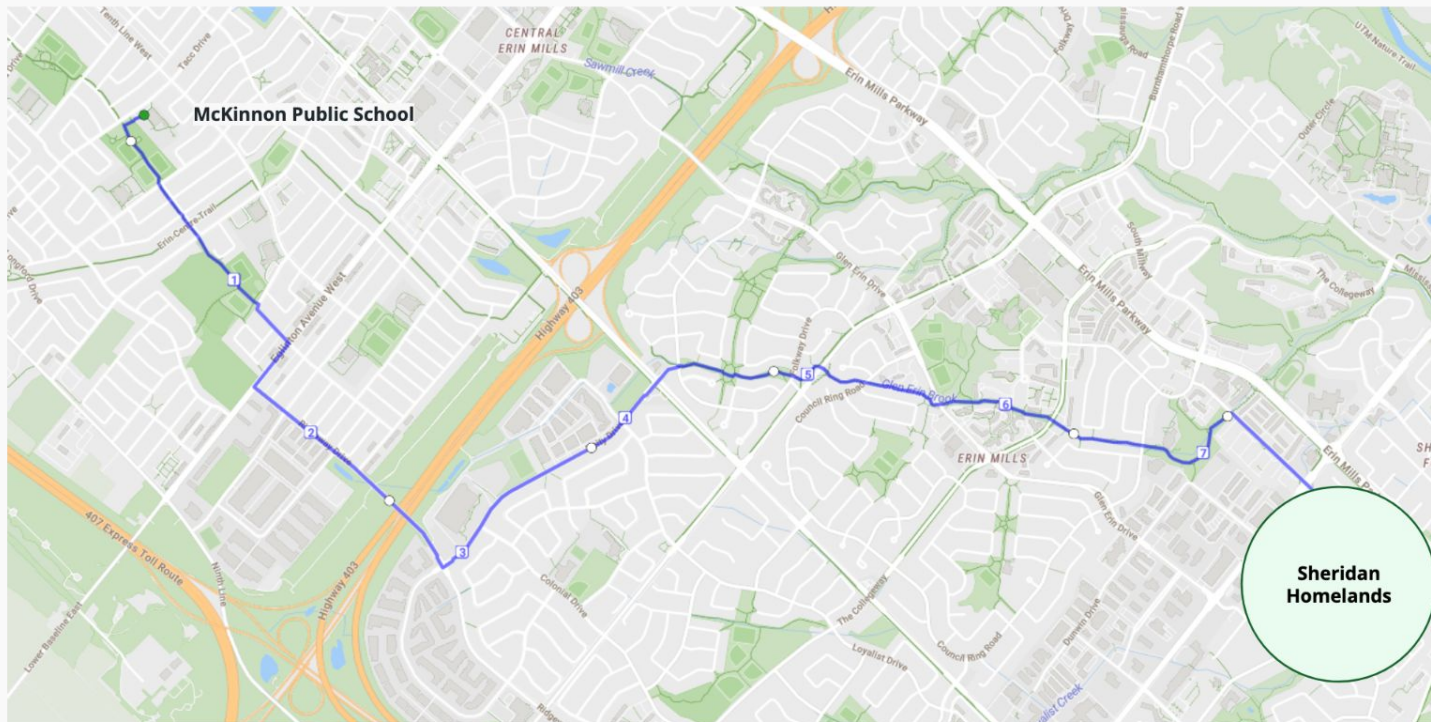
Early e-bike
adopter

Stored bike in
house /
classroom

~25 minutes

Crosses one
highway

Mostly multi-use
trails



Myrna's Commute

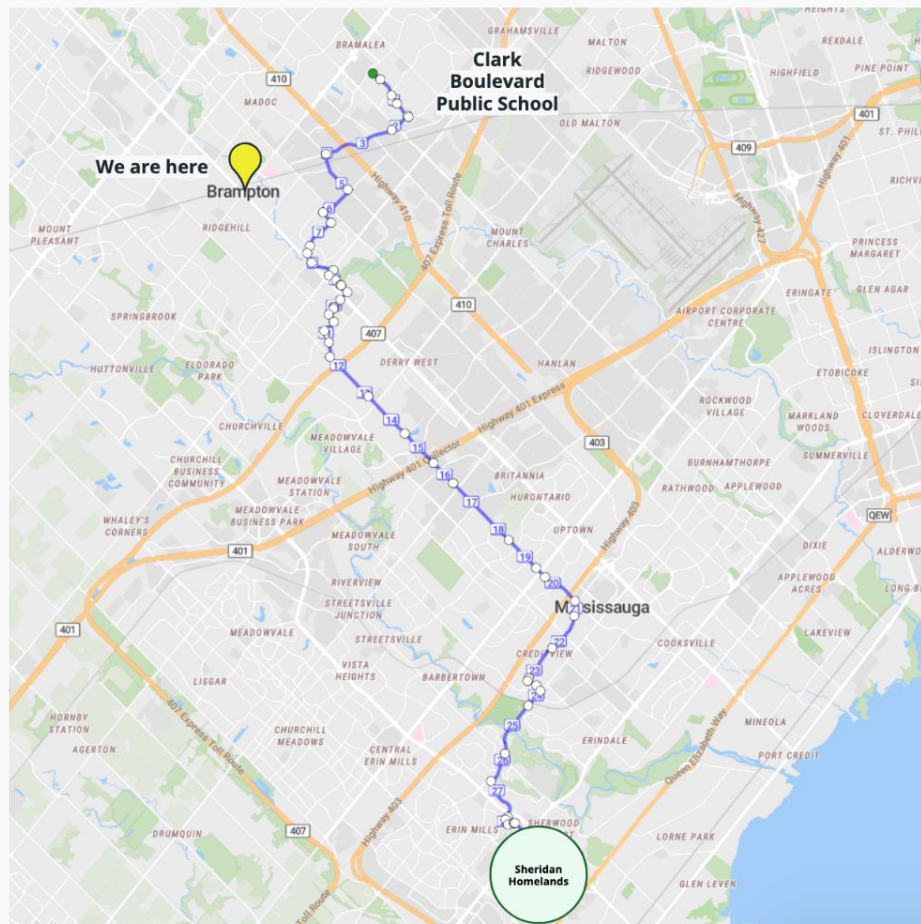
~1.5 hours

work → home trip only

Stored bike in house /
classroom or in teacher's
lounge

Crosses four highways and two
railway tracks

Strategic use of neighbourhood
roads, cut-throughs, and
multi-use trails



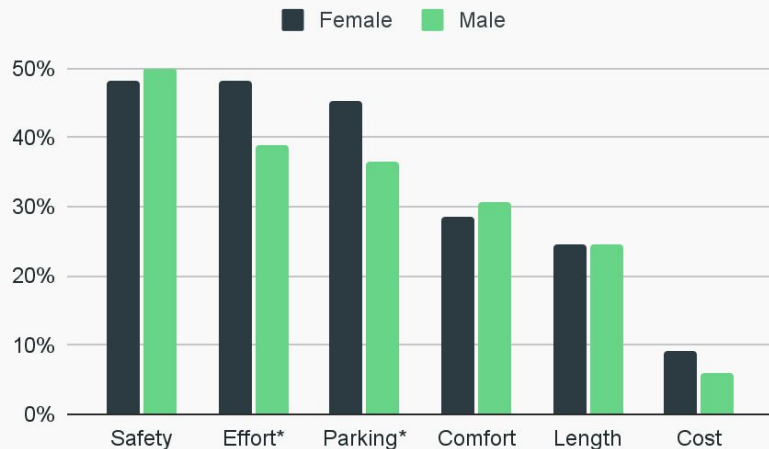


Why does bicycle parking matter?

Barriers to choosing cycling and impact of theft



Barriers to Cycling as cited by Potential Cyclists from McGill University

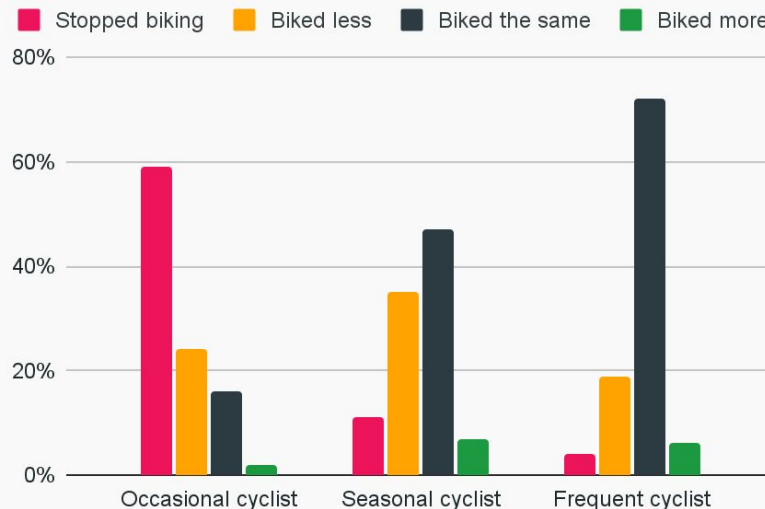


* = statistically significant difference

[Manaugh et al 2017](#)

[Overcoming barriers to cycling in a university setting](#)

Post-theft change in behaviour



[Cohen et al 2024](#)

[The impact of bicycle theft on ridership behavior](#)



4.5 BICYCLE PARKING

In order to foster greater use of a well-connected cycling network for more utilitarian trip purposes – as well as recreational ones - it is important that trip end destinations provide safe, comfortable, and convenient places to secure a bicycle. Bicycle parking is an essential component of a comprehensive active transportation network, and includes the following considerations:

- Identifying locations that are suitable for the installation of bicycle parking
- Selecting suitable types of bike parking racks depending on the parking need (long-term vs. short-term storage)
- Ensuring proper placement of the racks in the public right-of-way (including spacing between bike racks)
- Providing an appropriate number of spaces to provide for various types of development
- Considering other end-of-trip facilities such as bike repair stands, change and shower facilities, lockers, bench, etc.

Recommendation:

Prepare an inventory of bicycle parking located within the City that is available to the public, to identify gaps and to inform a future bike parking program. The inventory should include (but not be limited to) public spaces (roads, parks) recreation centres, civic buildings, schools, commercial properties, office space, etc.

Recommendation:

Update the City of Brampton parking zoning bylaw to require all new developments to include an appropriate supply of bicycle parking, taking guidance from the best practices outline within the ATMP.

Recommendation:

Incorporate the policies outlined in the ATMP relating to end-of-trip facilities into the development of the parking zoning by-law.

Brampton Active Transportation Master Plan



4.5 BICYCLE PARKING

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- **"Bicycle parking is an essential component of a comprehensive active transportation network"**
- Ensuring proper placement of the racks in the public right-of-way (including spacing between bike racks)
- Providing an appropriate number of spaces to provide for various types of development
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Recommendation:

Prepare an inventory of bicycle parking spaces. The inventory should include (but not be limited to) public spaces (roads, parks) recreation centres, civic buildings, schools, commercial properties, office space, etc.

Recommendation:

"require all new developments to include an appropriate supply of bicycle parking"

Recommendation:

"Incorporate... end-of-trip facilities into the... zoning by-law"



BikeSpace Tools

Reporting bicycle parking problems



7:28

AA bikespace.ca

BikeSpace Feedback

Where was the problem?

Pin the location

Map showing a location pin.

Back Next

7:28

AA bikespace.ca

BikeSpace Feedback

What were the issue(s)?

Bicycle parking is **not provided** ✓

Bicycle parking is **full**

Bicycle parking is **damaged**

A bicycle is **abandoned**

Something else

Next

7:30

AA bikespace.ca

Map showing various locations marked with green circles (IDs: 11, 44, 110, 1109, 7, 18, 38).

Latest Submissions

Thursday, February 26, 2026
last month • 11:37 PM ID: 1600
Parking damaged
Wanted to park for: several hours

Monday, February 23, 2026
last month • 8:11 AM ID: 1599
Parking damaged
Wanted to park for: less than an hour
Comments: Ping locate on one of the posts here

Back to Map

7:30

AA bikespace.ca

Map showing a location marked with a red circle (ID: 2).

Latest Submissions

Friday, November 14, 2025
last year • 12:00 PM ID: 1594
Parking damaged
Wanted to park for: several hours
Comments: Post rammed by a car and broken off; not replaced. Single remaining nearby bike lockup is often full (both sides).

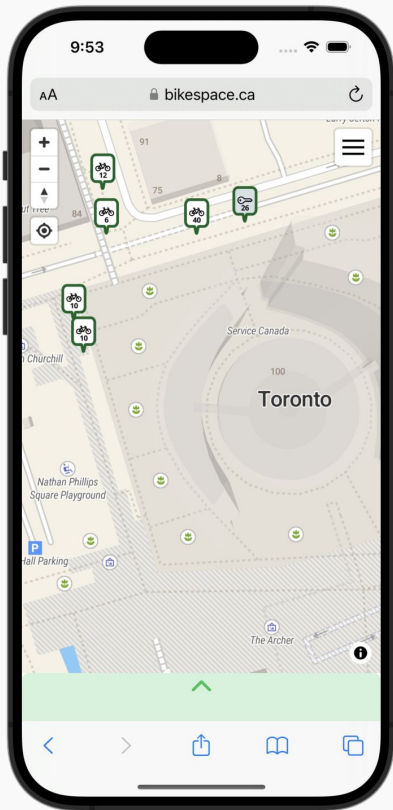
Saturday, November 8, 2025
last year • 9:37 AM ID: 1590

Back to Map

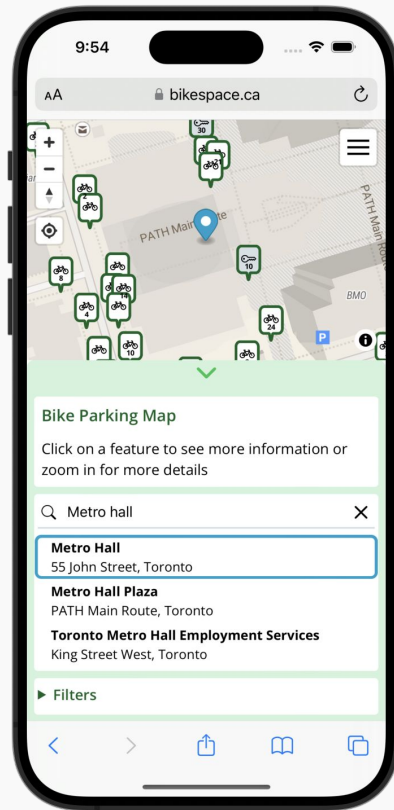
Finding bicycle parking in Toronto



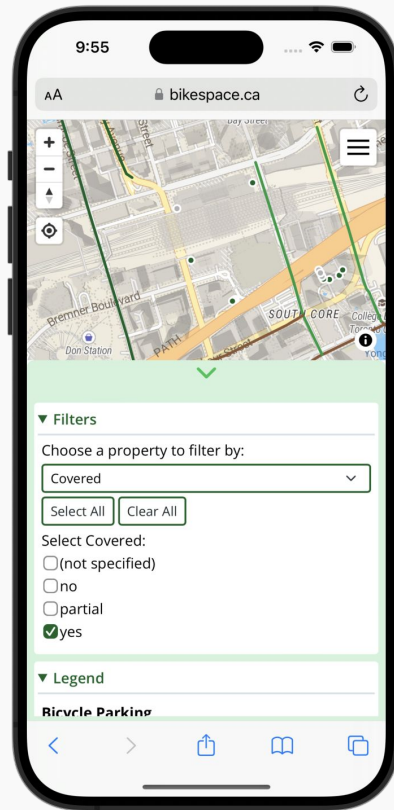
Find



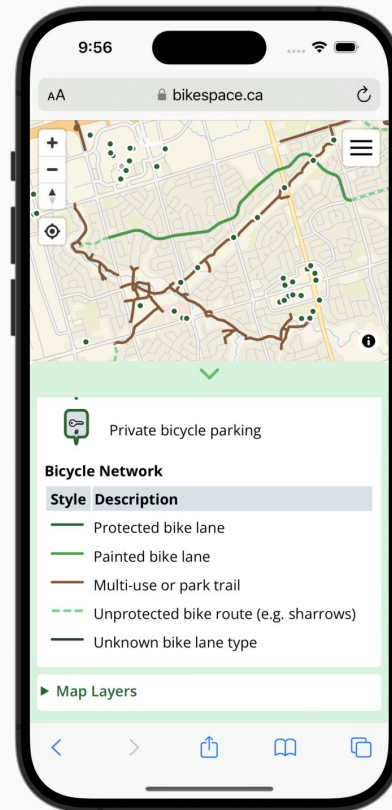
Search



Filter



Context



Rackfinder – OpenStreetMap Data



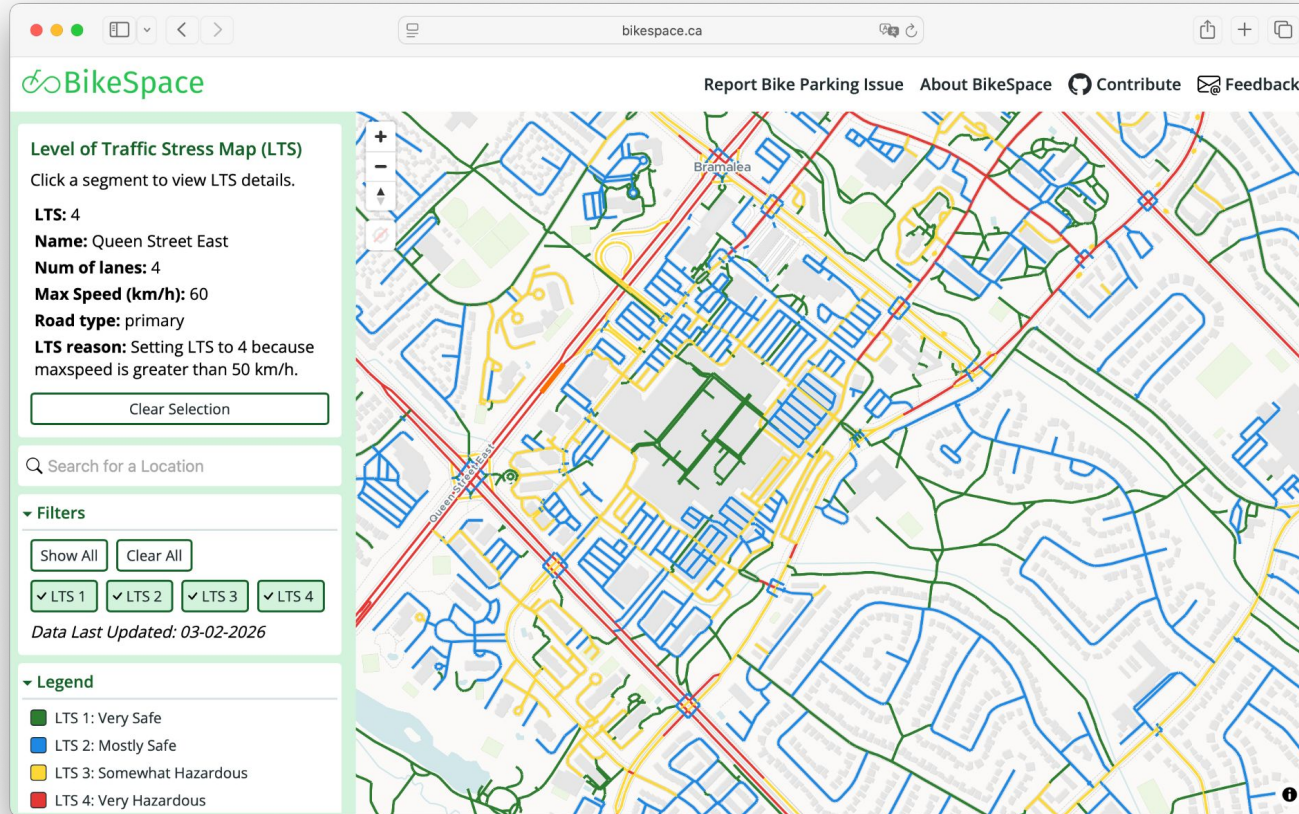
The screenshot shows the Rackfinder app interface. On the left, a sidebar titled "Nearby racks" displays 10 matching results. Each result includes a rack icon, a description, and a distance. A "Sort / filter" button is located at the top right of the list. The main area shows a map of a city street grid with several yellow rack icons placed at various intersections. The map includes street names like THOMAS ST, CHURCH ST E, UNION ST, MARKET ST, CHURCH ST W, QUEEN ST E, CHAPEL ST, JOHN ST, ELIZABETH ST, and MILL ST. A "Brampton Innovation District" label is also visible. The bottom of the app shows a navigation bar with a home icon, a user profile icon labeled "Not signed in", a "Sign in" button, and a settings gear icon. The browser address bar shows "rackfinder.app".

Rack Icon	Description	Distance
Unknown type	Unknown type	60 m
Stands	Stands	90 m
Stands Covered	Stands Covered	0.1 km
Rack 4 bicycles	Rack 4 bicycles	0.1 km
Rack Covered • 14 bicycles	Rack Covered • 14 bicycles	0.2 km
Stands Covered	Stands Covered	0.2 km
Stands Covered	Stands Covered	0.2 km
Unknown type	Unknown type	0.3 km
Rack Covered • 14 bicycles	Rack Covered • 14 bicycles	0.3 km
Unknown type	Unknown type	0.3 km



[alexwohlbruck/
rack-finder](https://github.com/alexwohlbruck/rack-finder)

Level of Traffic Stress





User Research

Alex Portolos

Timeline: Recent UX Research Initiatives at BikeSpace



Summer 2025

Recently Launched
BikeSpace Find Parking
Map ...

**Where do we go from
here?**

- We had a desire to:
 - **Better understand user pain-points** across their bike parking journey
 - **Evaluate usability** of new map
 - **Determine direction on next steps** for the new map

Timeline: Recent UX Research Initiatives at BikeSpace



Fall 2025

User Interviews and Usability Testing

(Lead by Alexei)



Heuristics Usability Analysis

(Lead by Malavika)

- **UX Research Initiatives:** BikeSpace conducted both user interviews and usability testing with 6 cyclists who bike in the GTA
 - Who are the cyclists of the GTA?
 - What is their journey and their challenges with respect to finding and using Bike Parking in the city?
- **Heuristic Usability Analysis:** BikeSpace quickly identified issues present on our tools by analyzing against established UX design principles.
 - It was a quick and low-cost way to uncover design problems.

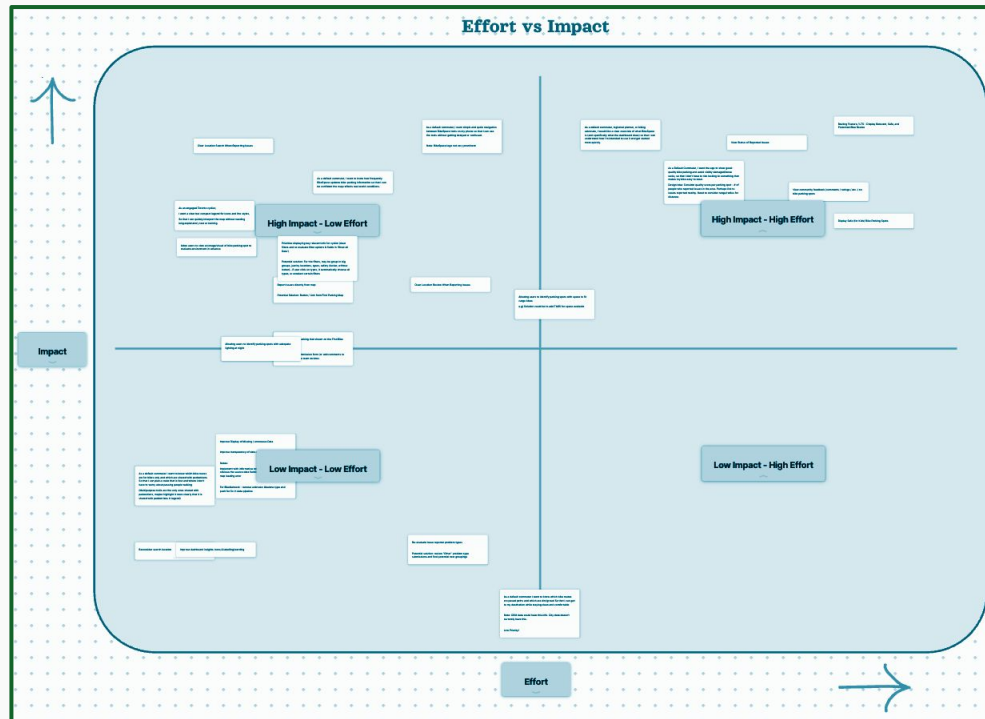
Timeline: Recent UX Research Initiatives at BikeSpace



2026 (Ongoing)

Identifying key focus areas and potential feature improvements to create a new roadmap for the BikeSpace Find Parking Map.

User Story & Prioritization: Converting these insights into user stories and prioritized them based on impact and effort.



Even a little UX research can go a long way



UX Research & Usability Testing:

- Learning from People's Lived Experiences
- Validate Assumptions Early On
- Usability Testing - Observe real users complete tasks to uncover issues
 - stay neutral, let them think aloud, observe, ask brief follow-ups

Advice for Teams with Limited Resources:

- Start Small
- Leverage Interview Guides
- Analyze Against Established Usability Principles

Don't forget about ethics!

- Exert Care Navigating Sensitive Topics
- Ask User's Permission for Recording / AI Analysis
- Anonymize Data Before Analysis



Who are the cyclists of the GTA?



The Default Commuter

P1, P3, P4, P5

- Generally bikes in downtown Toronto.
- Bikes for recreation and occasional commuting.
- Usually looks for bike parking at their destination.
- Doesn't have a lot of issues finding parking.
- Cares more about convenience and bike safety more than anything.



The Biking Advocate

P6

- Has more knowledge and interest around biking policy and biking infrastructure.
- May be a member of the local government or a bike-related organization.
- Has considered reaching out to the government to make changes to bike parking in the city.
- Thinks more about bike parking quality and capacity than the average person.



The Logistical Planner

P2

- Owns one or more non-standard bikes.
 - May occasionally bike with kids.
- Needs to plan bike parking due to specialized bike equipment and kids.
- Has more strict requirements over what bike parking is acceptable to them.

Logistical Planner User Journey



2. Choosing Bike, Route, and Parking Options

- Selects safest routes (smaller streets), prefers secure parking (underground or bike boxes), and adjusts choice of bike or locks based on location security.
- Considers Bike Share if practical, though cargo and distance can be limiting.



4. Unloading & Securing

- Prioritizes unloading kids safely, then handles cargo.
- Occasionally needs to coordinate with building staff for bike parking access, which adds time.



1. Advanced Planning (Night Before)

- Plans transport details the night before (ex. bike type, route, parking)
- Checks weather, researches area safety, and asks friends or online groups for advice.
- Uses Google Maps or websites to preview bike parking and routes.



3. Arrival & Parking Challenges

- Searches for spacious, safe, and convenient bike parking.
- Sometimes finds parking full or closed, requiring extra travel or heading back & switching to different transportation mode.
- May drop off kids first before parking farther away.



5. Reporting Issues

Reports damaged or stolen bikes and vandalized parking directly to building management.

Usability Heuristic Evaluation



Evaluated Against 10 Usability Principles:

1. Visibility of System Status
2. Match Between System and the Real World
3. User Control and Freedom
4. Consistency and Standards
5. Error Prevention
6. Recognition Rather than Recall
7. Flexibility and Efficiency of Use
8. Aesthetic and Minimalist Design
9. Help Users Recognize, Diagnose, and Recover from Errors
10. Help and Documentation

Recommendations Included:

- Several "Quick Wins"
- The need for a BikeSpace Design System

Button	Found on	Type/Use
Back	Page for issue reporting	Secondary action, allows users to return to prior screen
Cancel	Home page	Primary action, links to the dashboard
Cancel	Page for issue reporting	Form button, allows user to select one or more options from many
Issue page	Home page	Primary action, leads user to the reporting screen
Search Issue	Dashboard	Secondary action, allows user to filter between list or more options
Show All Data	Issue reporting page	Secondary action, allows users to select additional data
Save	Page for issue reporting	Primary action, allows user to proceed to the next step

Sample New Features (Part of Larger Roadmap)



**Open License
Street-Level Imagery**



**RentSafeTO Data on Bike
Parking in Apartments**



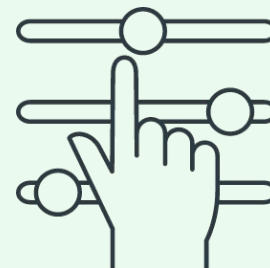
Bike Routes by Stress



**Cyclist can submit parking
data not shown on map**



**Community feedback
through reviews & ratings**



**Intuitive Filters &
Legends**



Thanks for listening!

